

EX.NO:7 IMPLEMENTATION OF A MULTI-THREAD PROGRAM

PROGRAM:

```
import java.util.Arrays;
import java.util.Random;
class SharedArray {
    int[] arr;
    SharedArray(int size) {
        arr = new int[size];
    }
    void generateArray() {
        Random r = new Random();
        for (int i = 0; i < arr.length; i++) {
            arr[i] = r.nextInt(100);
        }
        System.out.println("Generated Array: " + Arrays.toString(arr));
    }
    void ascendingSort() {
        int[] temp = arr.clone();
        Arrays.sort(temp);
        System.out.println("Ascending Order: " + Arrays.toString(temp));
    }
    void descendingSort() {
        int[] temp = arr.clone();
        Arrays.sort(temp);
        for (int i = 0; i < temp.length / 2; i++) {
            int t = temp[i];
            temp[i] = temp[temp.length - 1 - i];
            temp[temp.length - 1 - i] = t;
        }
        System.out.println("Descending Order: " + Arrays.toString(temp));
    }
}
class GenerateThread extends Thread {
    SharedArray a;

    GenerateThread(SharedArray a) {
        this.a = a;
    }
    public void run() {
        a.generateArray();
    }
}
```

```

class AscendingThread extends Thread {
    SharedArray a;
    AscendingThread(SharedArray a) {
        this.a = a;
    }
    public void run() {
        a.ascendingSort();
    }
}
class DescendingThread extends Thread {
    SharedArray a;
    DescendingThread(SharedArray a) {
        this.a = a;
    }
    public void run() {
        a.descendingSort();
    }
}
public class MultiThreadSort {
    public static void main(String[] args) throws InterruptedException {
        SharedArray a = new SharedArray(10);
        GenerateThread t1 = new GenerateThread(a);
        AscendingThread t2 = new AscendingThread(a);
        DescendingThread t3 = new DescendingThread(a);
        t1.start();
        t1.join();
        t2.start();
        t3.start();
        t2.join();
        t3.join();
        System.out.println("All threads completed.");
    }
}

```

OUTPUT:

Generated Array: [45, 12, 78, 34, 90, 23, 67, 11, 56, 30]

Ascending Order: [11, 12, 23, 30, 34, 45, 56, 67, 78, 90]

Descending Order: [90, 78, 67, 56, 45, 34, 30, 23, 12, 11]

All threads completed.